

Physics Informed Machine Learning for Cross Surface Droplet Impact Prediction

Testing on laser textured aluminium by holding out one surface at a time

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INTRODUCTION

1 Background and objective

Machine learning models for droplet spreading are usually validated on random splits of the data, so every test surface also appears in training [2]. Surface design needs the reverse: a reliable prediction for a surface that has not yet been made.

OBJECTIVE

Predict the maximum spreading ratio $\beta_{\max} = D_{\max}/D_0$ on laser textured surfaces excluded from training, and state how far each prediction can be trusted.

1,498

impacts on 12 textured surfaces

125

impacts on the smooth plate REF-H

1 to 160

mPa-s, water to 91 wt% glycerol

8 to 119

Weber number

9 to 4,435

Reynolds number

50 to 800

μm channel pitch, 6 or 25 μm deep

Experimental data from Može et al. [1], University of Ljubljana.

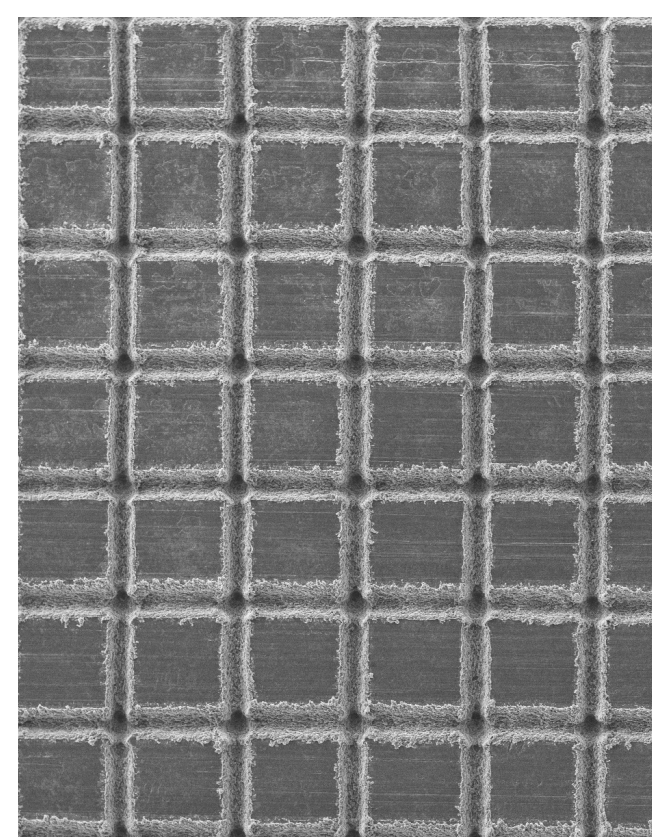
METHODS

2 Surface descriptors from SEM

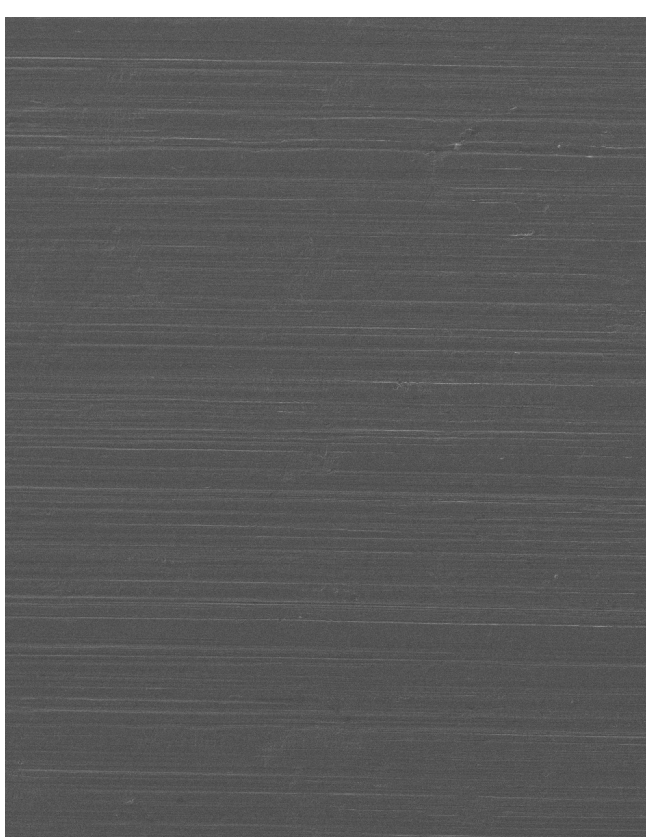
Between the laser tracks, the surface is flat and looks like the smooth plate. We measured the track width w on SEM images (about 45 μm for deep tracks and 30 μm for shallow ones) and used it to define two descriptors for each surface:

$$\varphi = (1 - w/s)^2 \quad V_{\text{tex}} = (1 - \varphi) h$$

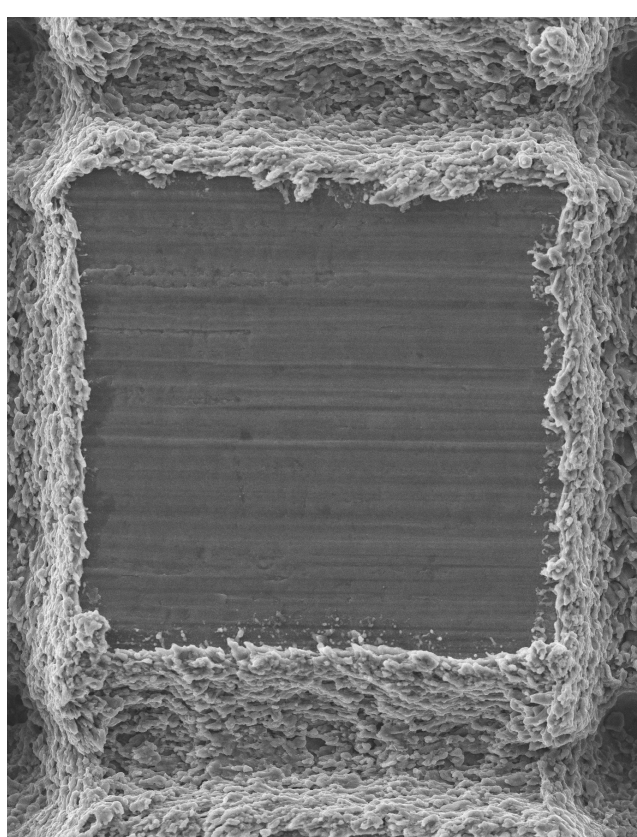
Here φ is the flat area fraction. It ranges from 0.01 (D50) to 0.93 (S800), and the smooth plate REF-H has $\varphi = 1$. Coding surfaces by channel spacing and depth instead shifted the REF-H error by up to 0.022. With φ , a 10 μm change in track width shifts it by 0.005 at most.



D200 textured, 43×

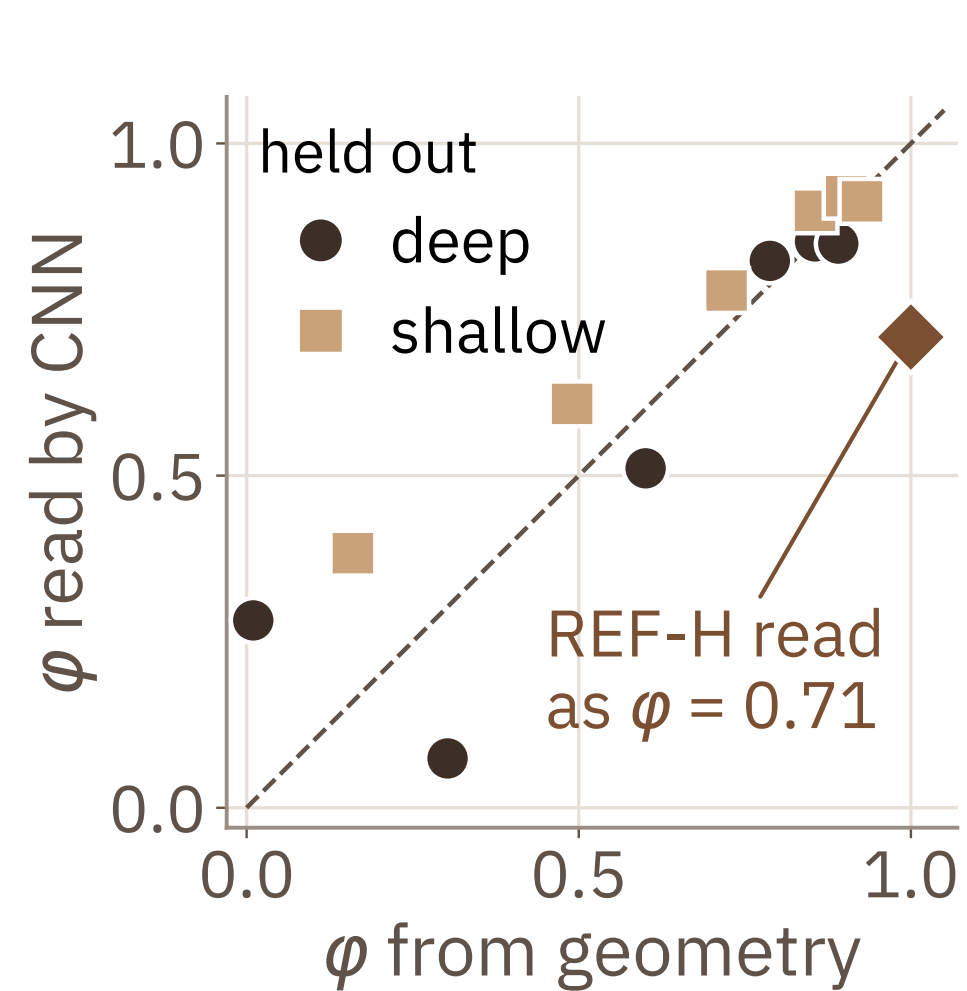


Smooth REF-H, 43×



D200 plateau, 350×

3 Reading descriptors directly from images



We trained a small CNN (9,070 parameters) to estimate φ and V_{tex} from three SEM images (43×, 100×, 350×) of a surface left out of training. Using these estimates, the GP error was 0.038 against 0.040, lower on 8 of 12 surfaces, but the difference is not significant. The CNN gives $\varphi = 0.71$ for REF-H, since the smoothest training surface has $\varphi = 0.93$.

4 Dimensionless inputs

$$Re = \frac{\rho V D_0}{\mu} \quad We = \frac{\rho V^2 D_0}{\sigma} \quad \beta_{\max} Re^{-1/5} = \frac{\sqrt{P}}{A + \sqrt{P}}$$

Our recomputed Re and We agree with the dataset to within 5×10^{-7} . The Ohnesorge number follows from Re and We , so we did not add it. The scaling law of Laan et al. [3], with $P = We Re^{-2/5}$, was developed for wettable surfaces. On this data it gives $R^2 = 0.71$, with an average error of 11% on held out surfaces.

5 Evaluation

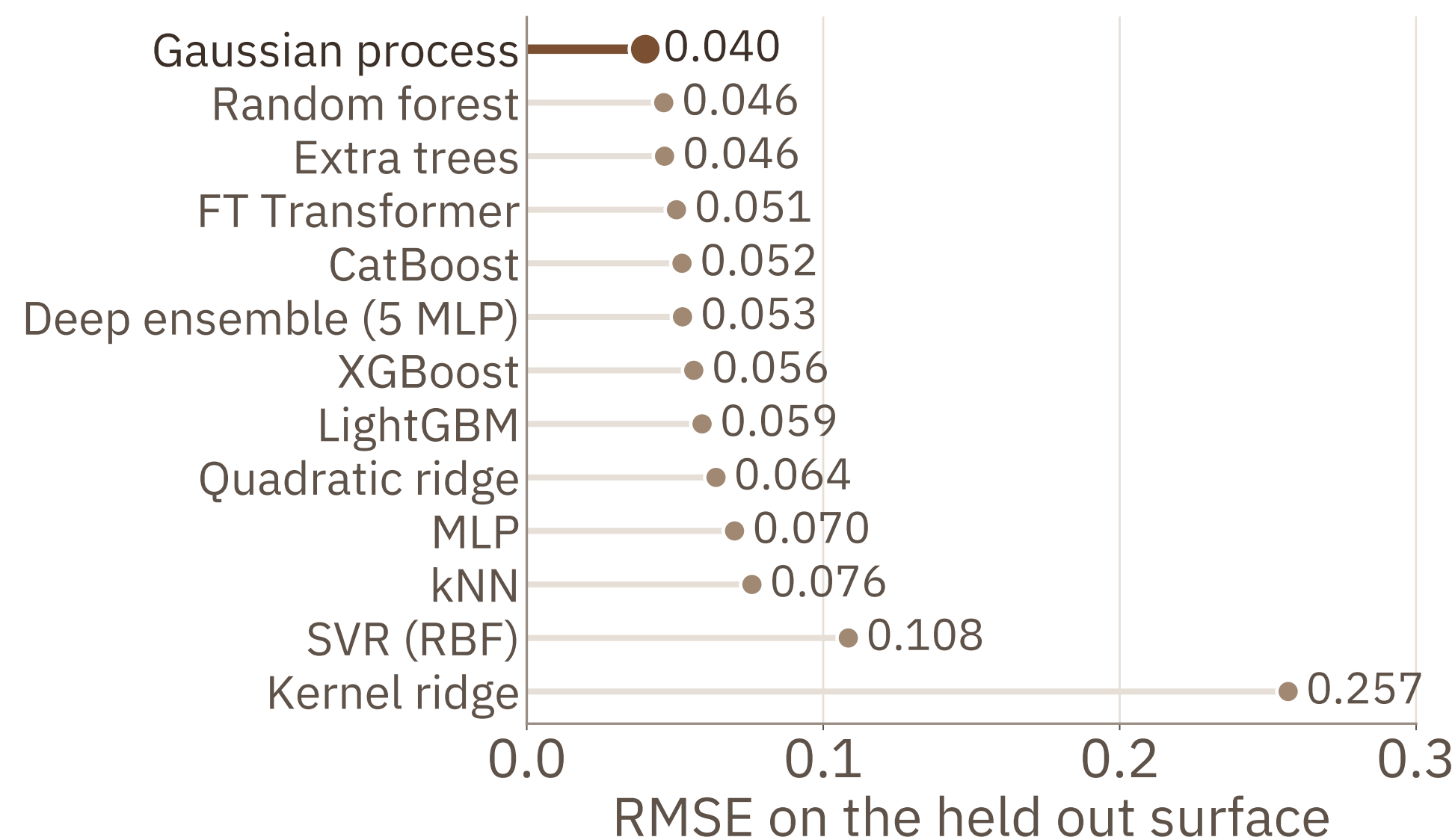
Every impact condition was repeated about five times. A random split would place repeats of the same condition in both training and test sets, so we used:

- 1 **Grouped 5 fold cross validation** over the 297 conditions, for surfaces seen in training.
- 2 **Leave one surface out** (12 folds), used to choose the model.
- 3 **REF-H kept aside** and never used for training or model choice.

Model inputs are $\ln Re$, $\ln We$, $\ln D_0$, φ and V_{tex} , and the target is $\ln \beta_{\max}$. We kept a change only if its 95% confidence interval, from 4,000 bootstrap resamples of whole surfaces, excluded zero.

RESULTS

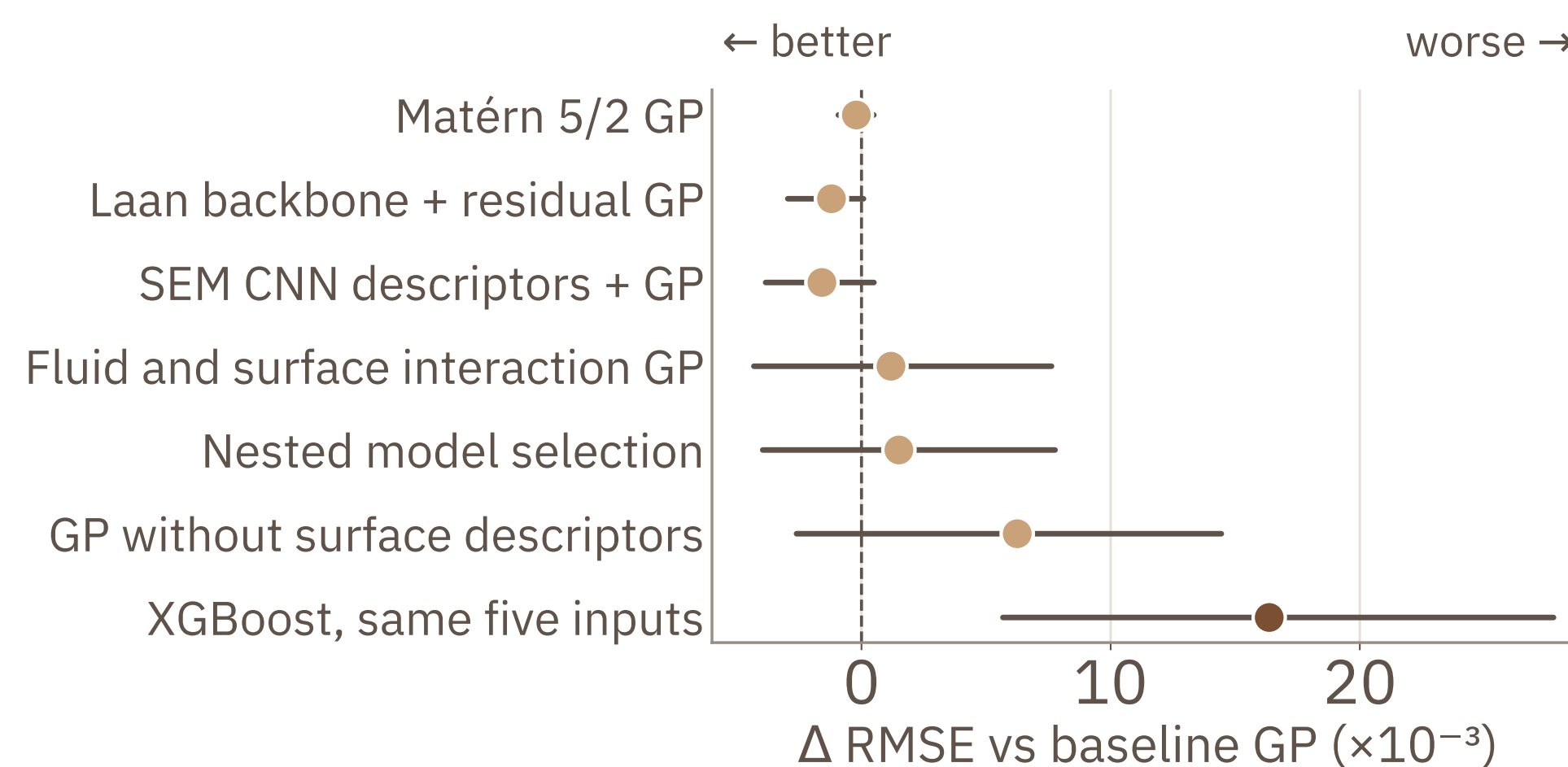
6 Comparison of 13 models



0.040

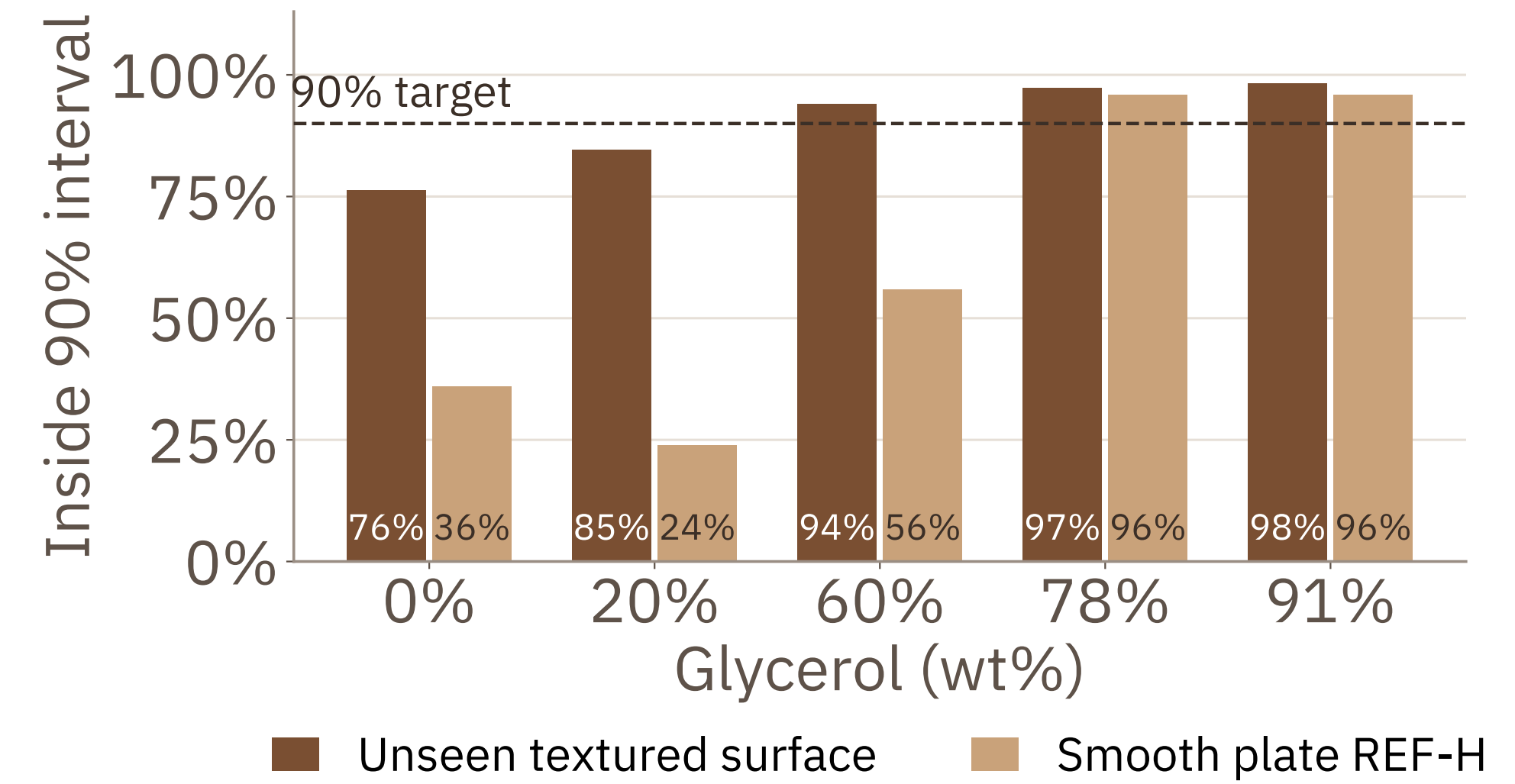
RMSE of the Gaussian process (Matérn 3/2 kernel) on held out surfaces, a mean error of 1.1%. XGBoost gives 0.056 (95% CI of the difference: 0.006 to 0.028). On surfaces seen in training, the GP gives 0.036.

7 Variants tested



Bars show 95% confidence intervals from paired bootstrap over surfaces. Adding a Laan law backbone, changing the kernel, nested model selection and CNN descriptors each changed the error by less than 0.002. XGBoost was the only significant difference, and it was worse. Removing φ and V_{tex} raised the error from 0.040 to 0.046. We therefore kept the plain GP.

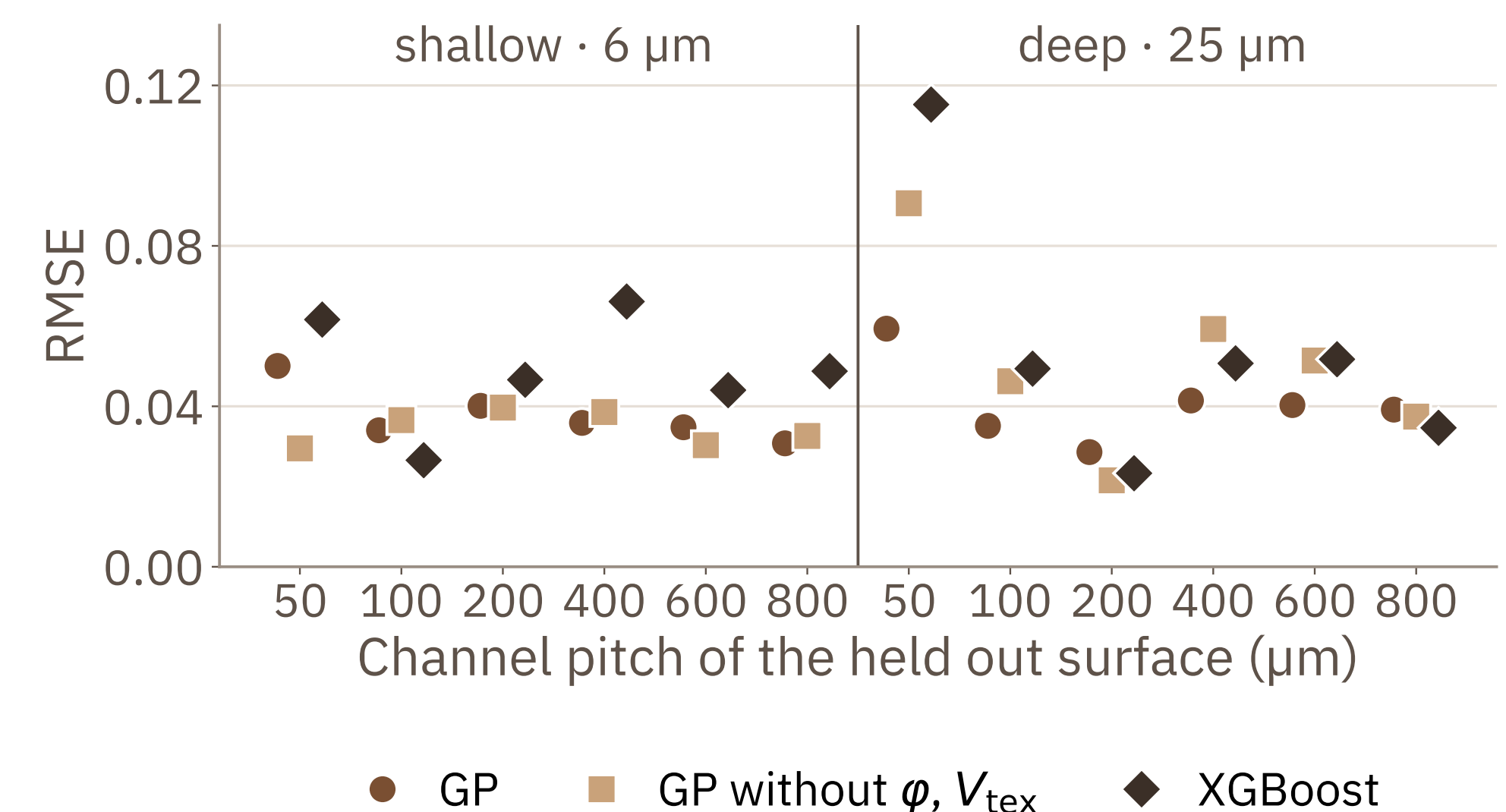
8 Prediction intervals



90.1%

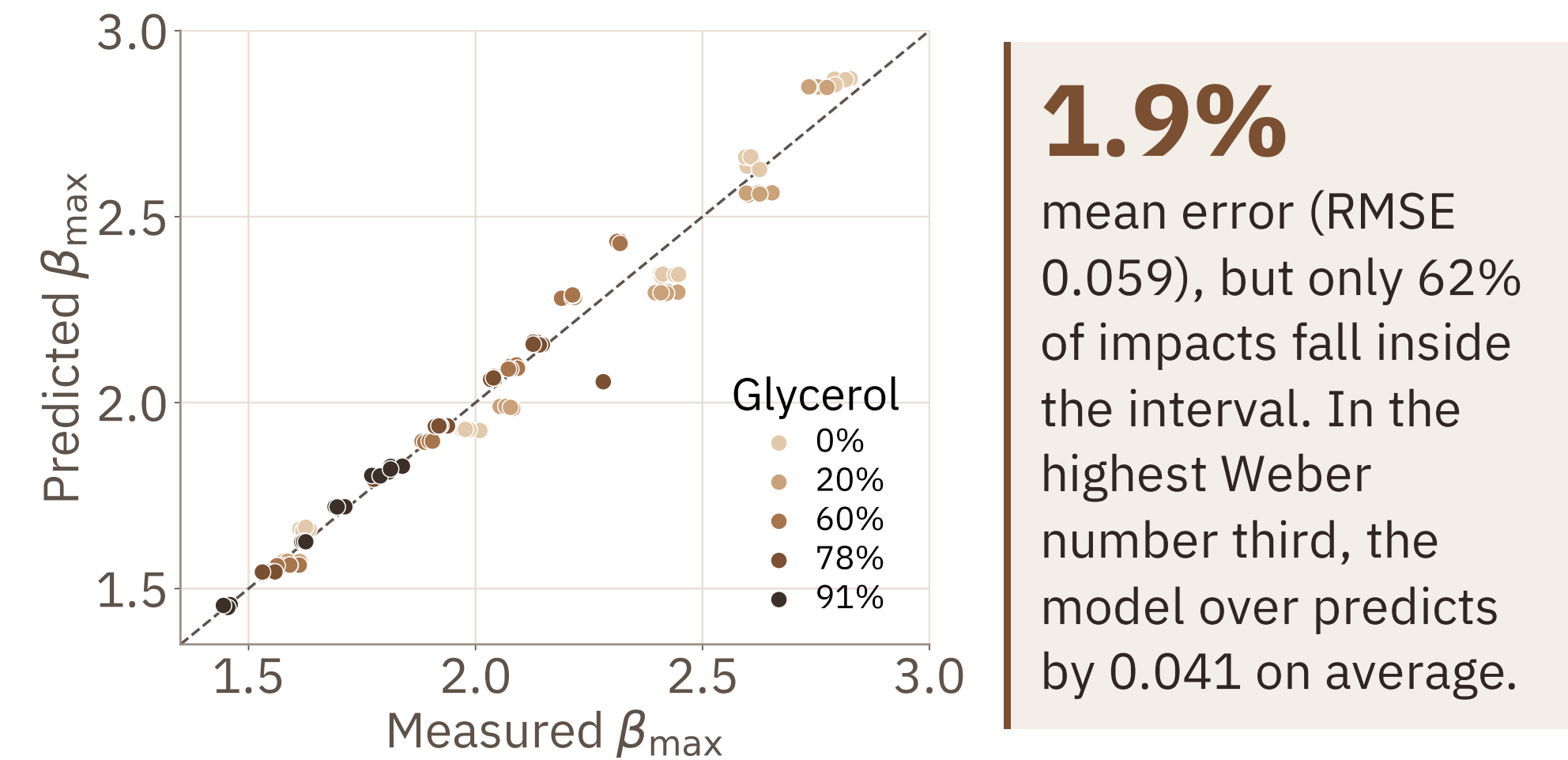
of impacts on held out surfaces fall inside the 90% interval (84 to 96% per surface). Coverage is 76% for water and 98% for 91 wt% glycerol. The out of range check flags D50 and S50.

9 Error for each held out surface



D50 ($\varphi = 0.01$, the edge of the range) has the largest error for every model. The descriptors lower it from 0.091 to 0.059 but slightly raise the error on S50 and D200. With 12 surfaces, this is suggestive only.

10 Smooth plate REF-H



1.9%

mean error (RMSE 0.059), but only 62% of impacts fall inside the interval. In the highest Weber number third, the model over predicts by 0.041 on average.

Most misses are for water and the 20% and 60% mixtures (24 to 56% coverage); viscous fluids reach 96%. REF-H is hydrophobic while the textures are superhydrophobic, and wettability is not a model input. Lower friction over trapped air may let droplets spread further on the textures [\[HYPOTHESIS\]](#)

FUTURE WORK

- Measure advancing and receding contact angles for each fluid and surface pair and use them as inputs.
- Make the noise level depend on the Ohnesorge number.
- Add new surfaces, with a test set fixed before any tuning.

REFERENCES

- [1] M. Može et al., Data in Brief 61 (2025) 111697; doi:10.17632/wsh8rxwd38.1
- [2] S. Jereb et al., Biomimetics 10 (2025) 357
- [3] N. Laan et al., Phys. Rev. Applied 2 (2014) 044018

CONCLUSIONS

- ✓ Testing on held out surfaces gives a more realistic error than random splits.
- ☰ The flat area fraction φ describes both textured and smooth surfaces, so the smooth plate fits in as $\varphi = 1$.

- 💧 A Gaussian process had the lowest error (0.040) of 13 models. A physics backbone and larger models did not improve it significantly.

- ⬆️ Intervals are well calibrated overall but not for every fluid, and they are too narrow on the smooth plate.